PostgreSQL Source Code Installation: Build PostgreSQL from Scratch

Wayne C.
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PostgreSQL is renowned for its reliability, performance, and extensibility — and one of its greatest strengths lies in being fully open source. Installing PostgreSQL from source offers developers and engineers deep control over the build, tuning, debugging, and even customization of the database engine.

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Linux distro, this article guides you through the **entire source-based installation process** on Linux.

🔹 Why Install from Source?

While PostgreSQL is available via package managers (APT, DNF, YUM), building from source gives you:

- The **latest version** before it hits package repos
- The ability to **customize compile-time options**
- Easier debugging or instrumentation
- Full control over **installation paths**
- Access to bleeding-edge features and patches

📅 Prerequisites

Before you begin, ensure your system has:

```
sudo apt update
sudo apt install build-essential libreadline-dev zlib1g-dev flex bison lib
```

Red Hat/CentOS:

```
sudo dnf groupinstall "Development Tools"
sudo dnf install readline-devel zlib-devel bison flex openssl-devel
```

📥 Step 1: Download PostgreSQL Source Code

You can always get the latest stable release from the official site:

```
wget https://ftp.postgresql.org/pub/source/v16.2/postgresql-16.2.tar.gz
tar -xzf postgresql-16.2.tar.gz
cd postgresql-16.2
```

⚙️ Step 2: Configure Build Options

This prepares the build environment. You can customize installation directories and enable optional features.

```
./configure --prefix=/opt/pgsql \
            --with-openssl \
            --with-libxml \
            --with-python
```

Key flags:

- `--prefix`: Target install path
- `--with-openssl`: Enable SSL support
- `--with-python`: For PL/Python procedural language
- `--enable-debug`: Enable debug symbols

🔨 Step 3: Build the Source

Now compile:

```
make -j$(nproc)
```

This will take a few minutes, depending on your hardware.

Then install:

```
sudo make install
```

PostgreSQL will now be installed under */opt/pgsql* (or your chosen prefix).

👥 Step 4: Initialize the Data Directory

Create a PostgreSQL user if not already done:

```
sudo useradd postgres
sudo mkdir /opt/pgsql/data
sudo chown postgres:postgres /opt/pgsql/data
```

Initialize the database:

```
sudo -u postgres /opt/pgsql/bin/initdb -D /opt/pgsql/data
```

▶️ Step 5: Start PostgreSQL

To start manually:

```
sudo -u postgres /opt/pgsql/bin/pg_ctl -D /opt/pgsql/data -l logfile start
```

To stop:

```
sudo -u postgres /opt/pgsql/bin/pg_ctl -D /opt/pgsql/data stop
```

To connect:

```
/opt/pgsql/bin/psql -U postgres
```

🧩 Step 6: Environment Variables (Optional)

Add PostgreSQL binaries to your PATH:

```
echo 'export PATH=/opt/pgsql/bin:$PATH' >> ~/.bashrc
source ~/.bashrc
```

🔄 Step 7: Managing as a Service (Optional)

You can wrap PostgreSQL in a systemd service:

```
sudo nano /etc/systemd/system/postgresql-source.service
```

```
[Unit]
Description=PostgreSQL from Source
After=network.target

[Service]
Type=forking
User=postgres
ExecStart=/opt/pgsql/bin/pg_ctl -D /opt/pgsql/data -l /opt/pgsql/logfile s
ExecStop=/opt/pgsql/bin/pg_ctl -D /opt/pgsql/data stop
RestartSec=5sec

[Install]
WantedBy=multi-user.target
```

Then:

```
sudo systemctl daemon-reexec
sudo systemctl enable --now postgresql-source
```

✔️ Step 8: Test Your Build

To ensure everything's working:

```
/opt/pgsql/bin/psql -U postgres -c "SELECT version();"
```

You should see your compiled version with the enabled features.

🧬 Advanced: Build and Enable Extensions

Some extensions require source builds as well:

```
cd contrib
make
sudo make install
```

Then in psql:

```
CREATE EXTENSION pg_stat_statements;
```

🐳 When Should You Build from Source?

✅ Ideal for:

- Developers modifying PostgreSQL internals
- DBAs needing bleeding-edge features
- Building reproducible binaries
- Systems where distro-provided packages are outdated

❌ Avoid if:

- You prefer easy upgrades
- You're deploying many instances (use Docker or packages instead)
- You need managed support (use RDS, Cloud SQL, etc.)

📋 Summary Cheat Sheet

🚀 Wrap-Up

Installing PostgreSQL from source is straightforward if you're comfortable with Linux command-line tools and build systems. It's a great way to:

- Explore PostgreSQL's internals
- Optimize builds for your hardware
- Customize features without bloat

Once you're familiar, you can even automate the build with scripting or use it in containerized workflows.

#postgresql

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